

01

VORTEX GRIP

Braided over/under helical straps for dynamic multi-point contact.

46.5 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	37.8 × 31.7 mm
Peak texture height	2.85 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.47 cm³ / 318 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		46.5 g
PETG	1.27 g/cm³		47.6 g
TPU	1.21 g/cm³		45.3 g
ABS	1.04 g/cm³		39.0 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.6 m (100 % flow)

DESIGN INTENT

Sixteen straps wound in opposite directions cross in an over/under braid, leaving dark valleys between them. Every crossing gives the hand an edge to bite on, so the grip works from any hand position on the handle.

Straps rise no faster than 45 degrees, so no supports are needed.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 8 right-hand + 8 left-hand straps on a 36 degree helix
- Over/under weave: the strap on top alternates at every crossing
- Valleys drop to about the 1.2 mm base wall, which keeps the mass low

FILES

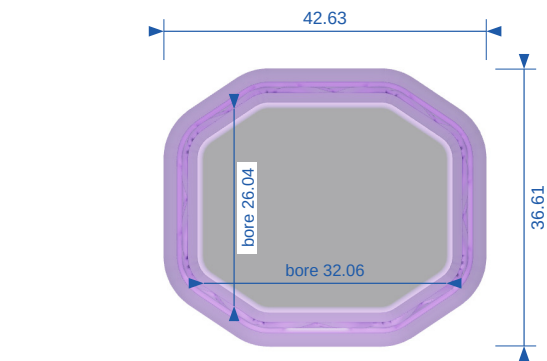
stl/01_vortex_grip.stl	19.9 MB · 398,860 triangles
3mf/01_vortex_grip.3mf	4.5 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

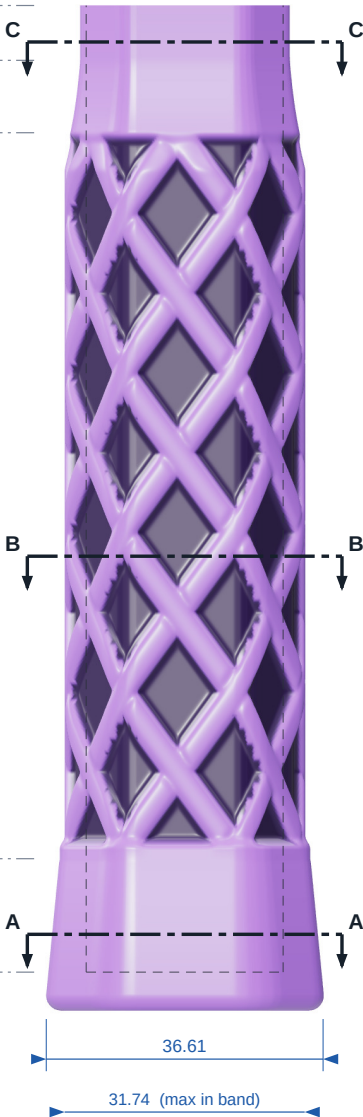
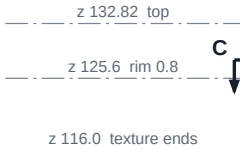
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	37.76 × 31.74
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.85
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

VORTEX GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

46.5 g

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

37.47 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

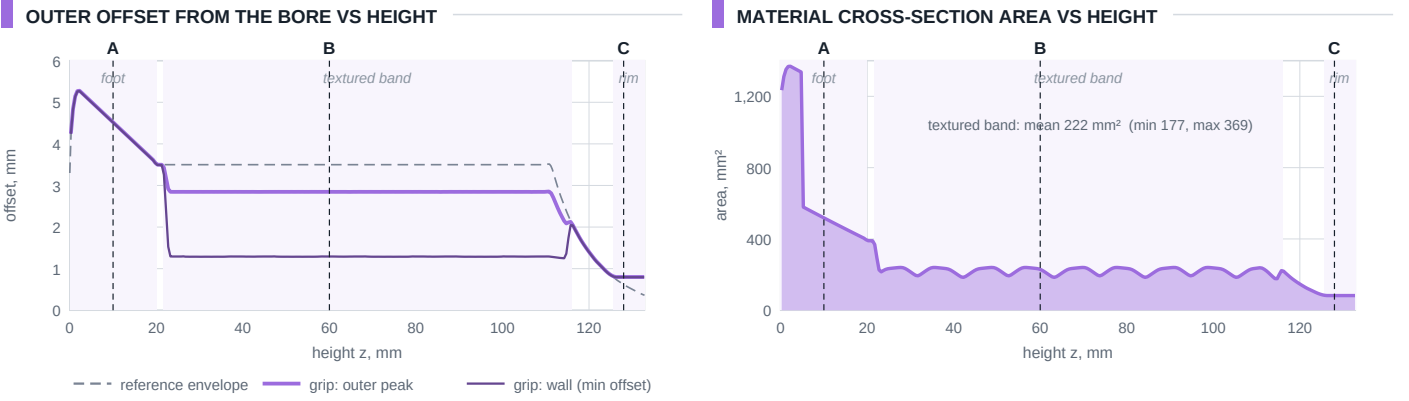
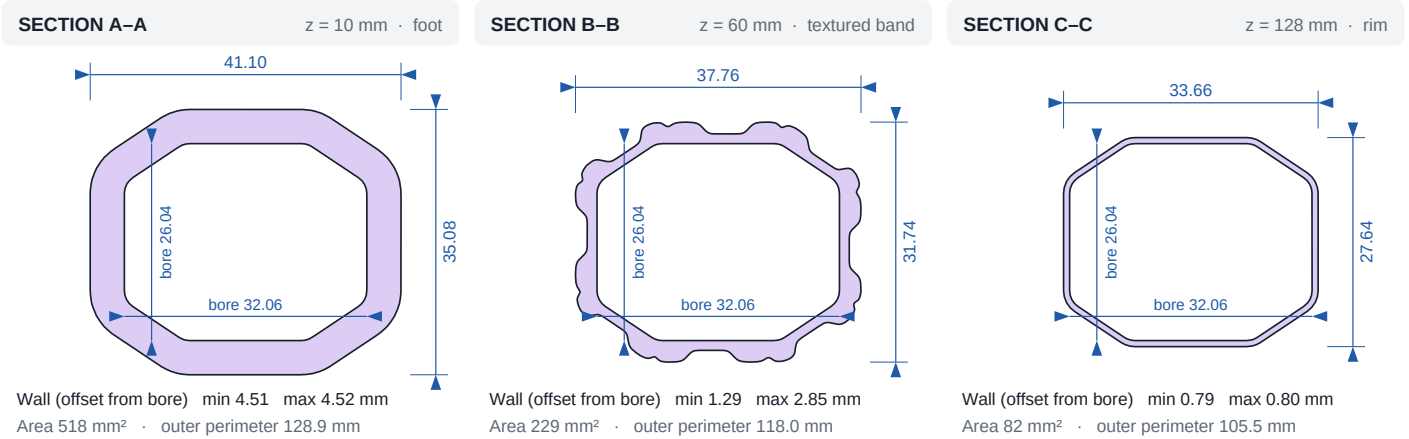
318 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	37.76 × 31.74 mm
Circumference in band	113.2 – 118.0 mm
Volume / surface area	37.47 cm ³ / 317.5 cm ²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.29 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			2.85 mm

OUTER DIMENSIONS BY HEIGHT

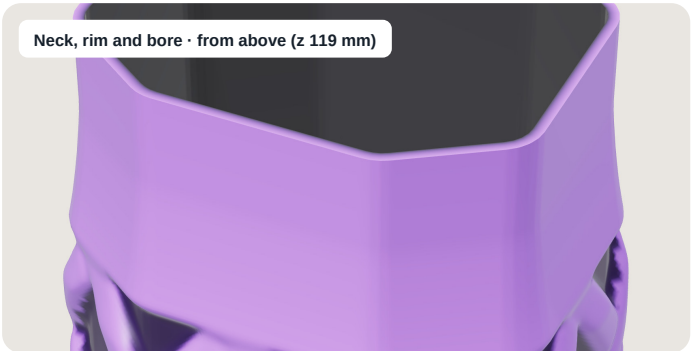
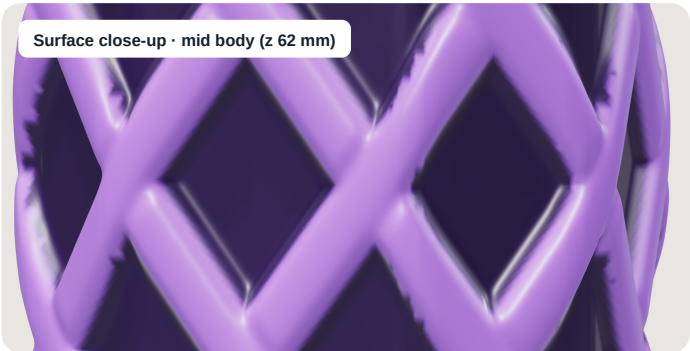
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.76	31.74	114.5	215	1.29	2.85
45.0	37.76	31.74	118.0	225	1.29	2.85
60.0	37.76	31.74	118.0	229	1.29	2.85
75.0	37.76	31.74	114.1	209	1.29	2.85
90.0	37.76	31.74	117.1	238	1.29	2.85
105.0	37.75	31.74	117.1	187	1.29	2.85
116.0	36.24	30.22	113.4	221	2.04	2.09
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

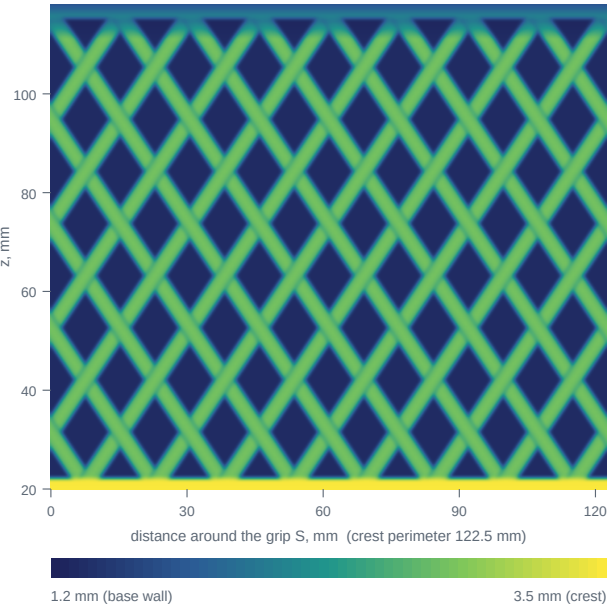
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Straps	8 + 8, counter-wound
Helix angle	36 deg from vertical
Strap pitch (around)	15.31 mm
Strap width	6.1 mm around / 5.0 mm across
Strap top height	2.35 – 2.86 mm above bore
Valley floor	1.29 mm above bore
Over/under step	0.37 mm

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none